

Maag

Modular Agentic Autonomous Greenhouse

12 steps · print files · wiring

Herb Snip assembly instructions, Rev A. The cut-and-hold tool for the myCobot 280.

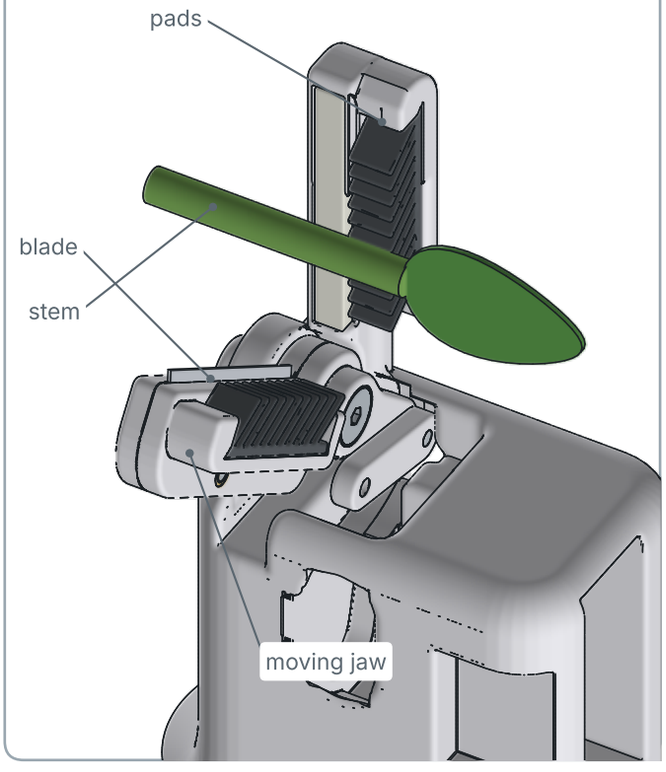
Design 0.7.0. The Maag cover render, the same on every booklet. Every drawing inside comes from snip_cad.py, Intel's D405 mesh and Elephant Robotics' arm meshes.

How the Herb Snip works

One servo closes one jaw. The pads take hold of the leaf's side of the stem, then the blade cuts.

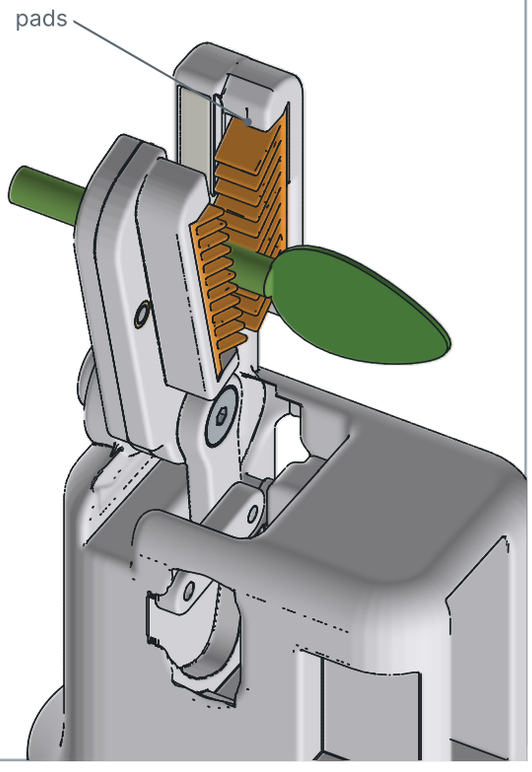
1 Open

the stem goes between the jaws,
the leaf on the servo side



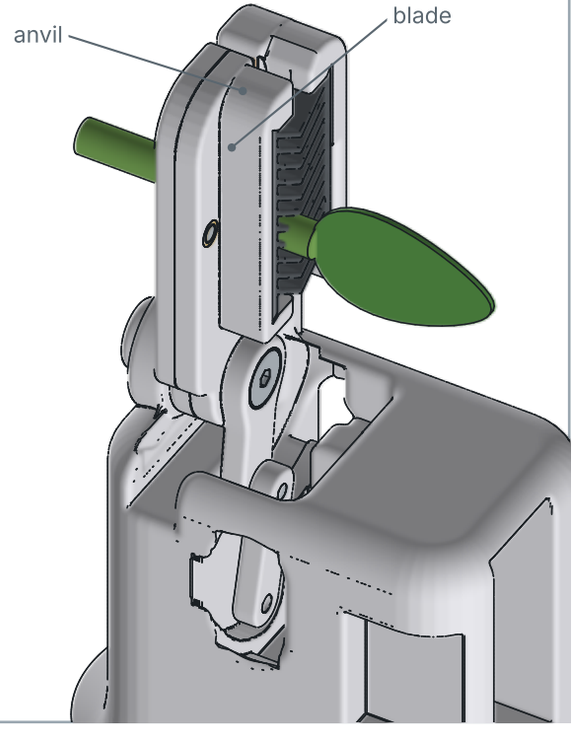
2 The pads close first

they hold the leaf's side of the
stem before the blade reaches it



3 Cut, and still held

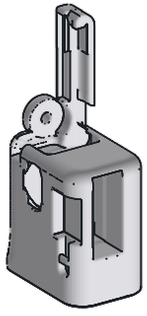
the blade meets the anvil; the
leaf stays in the pads



The servo turns a crank, the crank pulls a link, the link closes the moving jaw: 34 N at the middle of the blade through a 4.5 mm stem, with the servo held to 70 % of its stall torque. The pads are the TPU combs beside the blade, on the servo side. Present the plant so the part you want is on that side: it is the part the tool keeps. The other side is the plant, and stays behind.

Herb Snip · printed parts

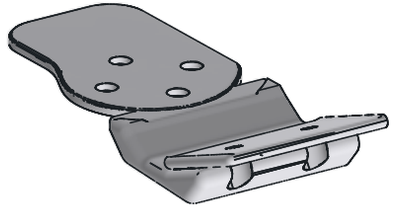
Seven parts in white PETG and a TPU pad printed twice: 73 g printed for the tool.



Body

49.2 g · 77.41 cm³

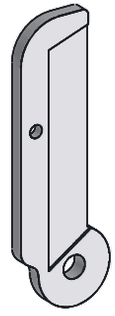
one housing: the collar above the band, the servo's pocket, the regulator's bay, the fixed jaw



Camera ring

15.7 g · 16.45 cm³

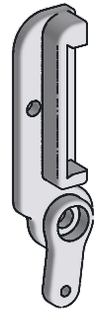
the collar's foot and the D405's chin; between the flange and the body, on the same four pins



Jaw A

2.0 g · 1.88 cm³

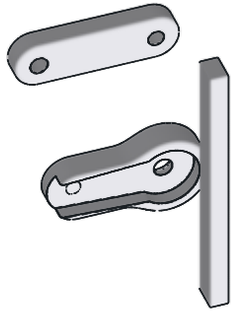
camera side; the blade's pocket, the clamp screw's countersink



Jaw B

3.2 g · 2.98 cm³

servo side; pad rib, tail lug, the clamp insert, the pivot's counterbore



Link, crank, anvil

1.1 g · 1.02 cm³

crank to jaw tail, the crank on the horn; the anvil is the strip the blade closes on, replace when scored deep



Pad, two

0.7 g · 0.58 cm³ each

TPU 95A: a comb of 11 fins on a dovetail; one on each jaw

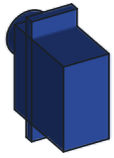
Print settings

Material	white PETG, like the arm; PLA creeps under 40 N
Pads	TPU 95A, solid, 20 mm/s, lying on their side
Layers	0.20 mm, 0.4 nozzle
Walls	4; small parts solid
Infill	body and ring 25 %
Supports	none; bridging on; the ring a 5 mm brim

snip/print/*.stl, already lying as they print

Herb Snip · purchased parts

Everything else on the tool. The D405 and the arm's own LEGO pins may already be in the box.



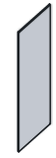
TowerPro MG90S servo, metal gear

1×



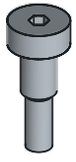
Pololu D24V10F6, 6 V 1 A step-down

1×



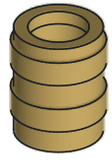
OLFA AB 9 mm blade, a 34 mm piece

1×



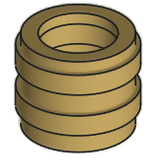
M3 shoulder screw, 4 × 6 · pivot

1×



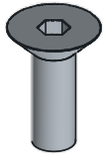
M3 heat-set insert, 5.7 long, 4.6 OD

1×



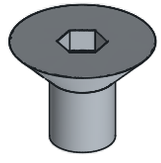
M2 heat-set insert, 3 long, 3.6 OD

1×



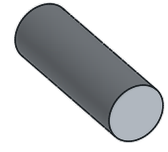
M2 × 6 countersunk · blade, crank

2×



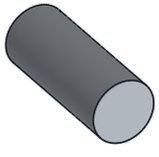
M3 × 5 countersunk · camera

2×



dowel pin 2 × 6, ISO 8734

1×



dowel pin 2 × 5, ISO 8734

1×



LEGO Technic pin with friction, 2780

4×

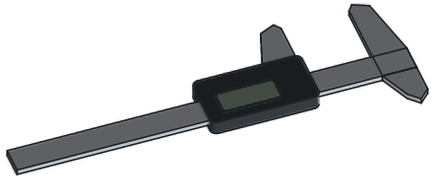
Also: 3-wire servo lead to the arm's end connector (2.54 Dupont), a red and black 22 AWG pair from the 12 V wrist line, 2 small zip ties, heat-shrink. Already on the arm: the D405 and its high-flex USB-C cable (BOM G056, G057).

Tools

A short list: the Herb Snip is small, and every screw is M2 or M3.



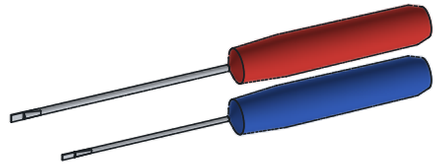
Hex keys
1.3 · 2 mm
steps: 1.3: the two M2 countersunk 2: the pivot and the camera screws



Calipers
150 mm, digital
steps: the blade piece; the horn's fit, 2



Multimeter
DC volts
steps: 6.0 V at the regulator before the servo goes on, 8



Small screwdrivers
Phillips 1 and flat 2.5
steps: the servo's tab screws, 7

`flat-nose pliers · the blade, 3`

`cut-resistant gloves · the blade, 3`

`soldering iron · 3, 4, 8`

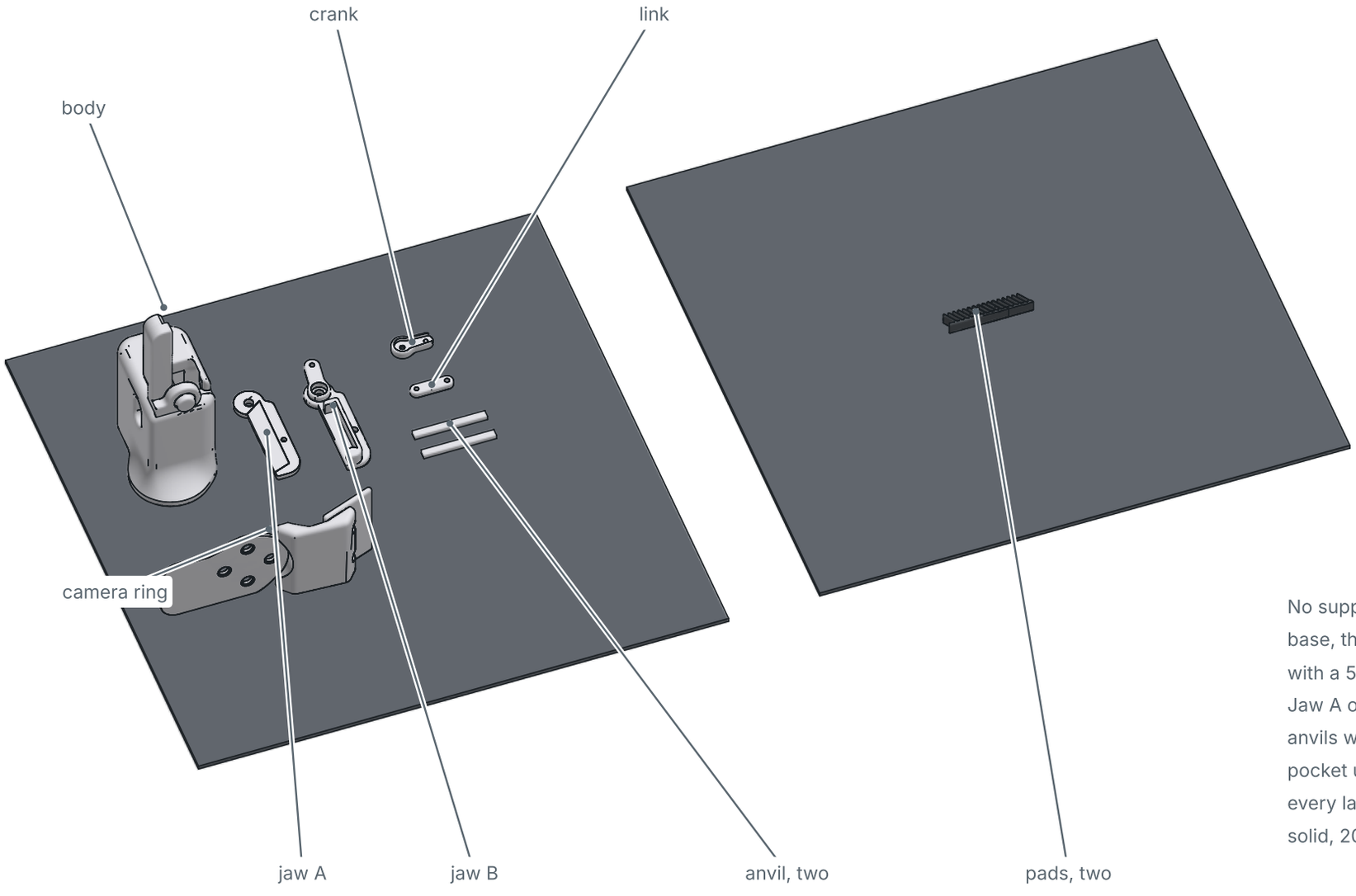
`wire stripper · 8`

`servo tester, or the arm · 2, 6`

No drill, no tap: every hole is printed to size. The two dowels press in with the flat of the pliers, the two inserts with the soldering iron at about 230 °C.

1 Print the parts

Each file already lies the way it prints. One plate in PETG, then the pads in TPU.

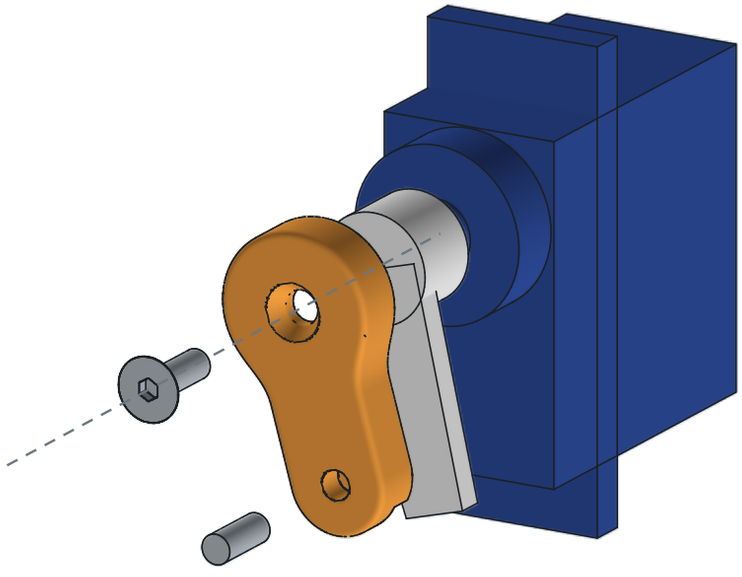


No supports. The body stands on its flat base, the camera ring on its flat side with a 5 mm brim (it stands 42 tall). Jaw A outer face down, jaw B and the anvils working faces down, the crank horn pocket up. The pads lie on their side, so every layer is the whole comb: TPU 95A, solid, 20 mm/s, a plate of their own.

A 256 × 256 bed (the Bambu P2S's). The anvil face and both jaw faces print against the plate: those are the faces that must be flat. Check the crank fits your servo's horn before anything else is printed (step 2).

2 Fit the crank to your servo's horn

TowerPro publishes no drawing of the horn. Check it before going on.



- 1× single-arm horn, from the servo's bag
- 1× crank (printed)
- 1× M2 × 6 countersunk
- 1× dowel pin 2 × 5, pressed into the crank

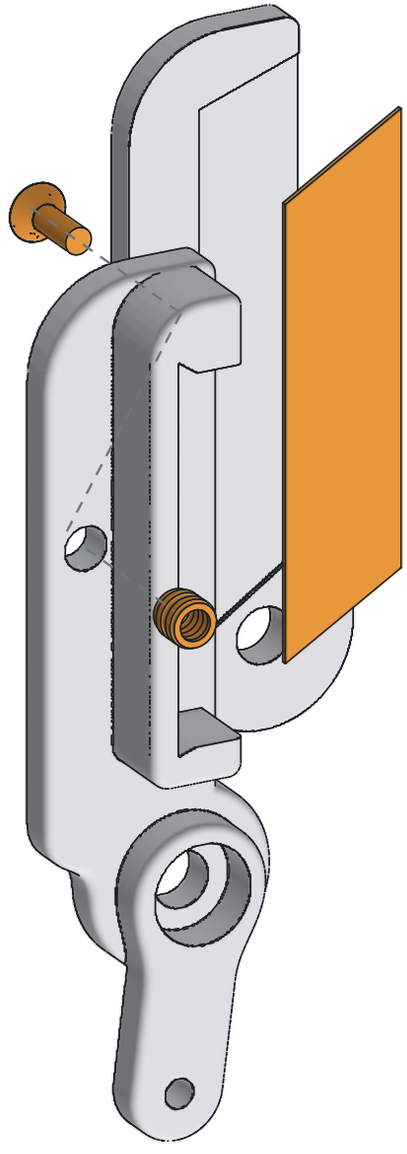
1.5 mm hex · flat pliers

The horn's arm drops into the crank's pocket and the crank sits flat on the horn's face. If the arm will not drop in, or the crank rocks, measure the horn and change HORN in snip_cad.py: the crank reprints in minutes.

Press the 2 × 5 dowel into the crank's small hole until it stands 3 proud on the horn side. Do not fit the crank to the servo yet: it goes on in step 6, at the servo's closed position, so the jaw closes where the servo stops.

3 The blade between the jaw halves

One insert, one blade, one screw. Snap a piece with about 34 of cutting edge. Gloves on.



- 1× OLFA AB 9 mm blade, a 34 mm piece
- 1× M2 heat-set insert, into jaw B
- 1× M2 × 6 countersunk, from jaw A

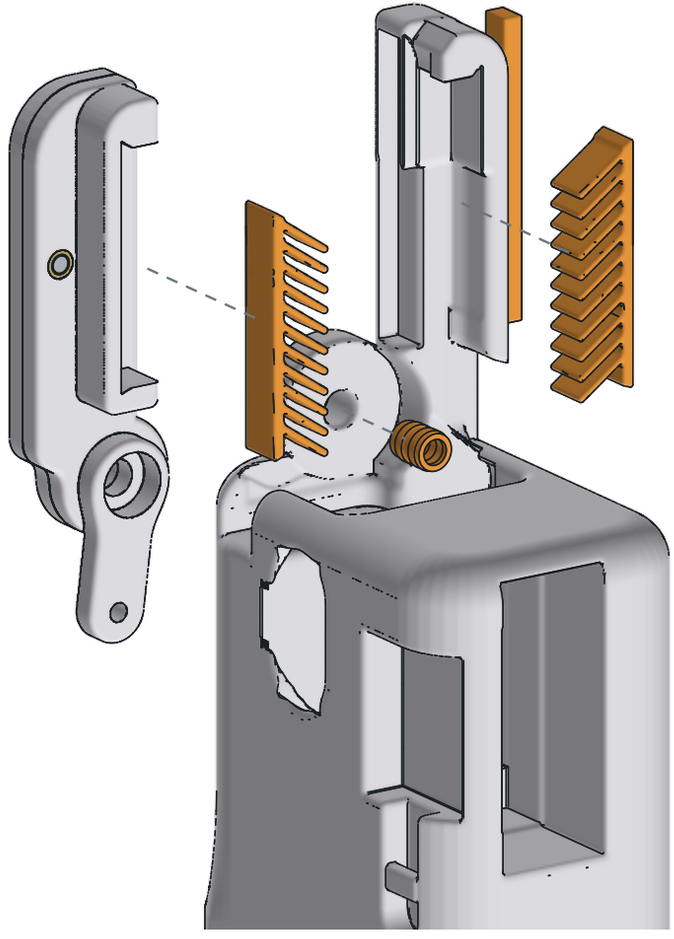
iron · 1.3 mm hex · pliers

First the insert: into jaw B from its outer face, the iron at about 230 °C, until it sits flush. Then the blade: grip it in the pliers at a score line and bend away from you. The piece's ends are the score lines, so it is a parallelogram: the pocket's angled ends take any OLFA piece of 30 to 35. Cutting edge toward the jaw's straight inner edge, sticking out 1.5 past it.

Lay the blade in jaw A's pocket, bevel down. Jaw B on top, faces together, and the screw from jaw A's side: its head sinks flush. The pocket is 0.08 shallower than the blade, so the one screw pinches it: snug, no more. The thread is brass, so a blade can be changed as often as it needs to be.

4 Pads, anvil and the pivot insert

The two TPU pads slide in from the servo side. The anvil and an insert go into the body.



- 2× pad (printed in TPU 95A)
- 1× anvil (printed)
- 1× M3 heat-set insert, into the cheek

`soldering iron`

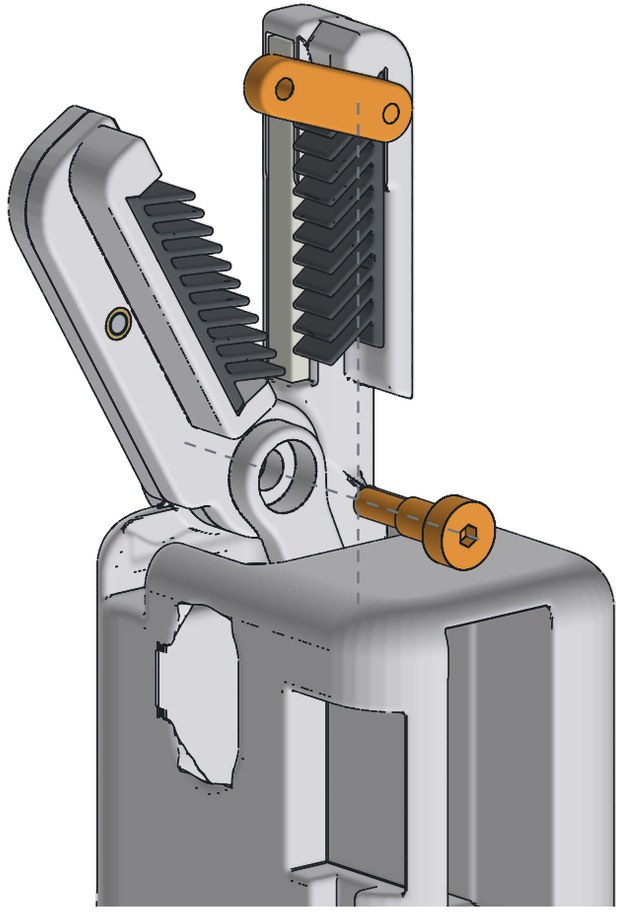
The insert goes in from the jaw side of the cheek, flush: the pivot's shoulder bears on it. The deck is 4 under its axis: an iron under 8 thick for its first 50 sets it square.

Each pad slides into its rib until it stops on the jaw plate, fins toward the other jaw. The dovetail is cut one way only: if a pad will not go in, turn it end for end. The anvil slides into its pocket, smooth face toward the blade; a drop of thin glue keeps it.

The fins stand 2.2 proud of the ribs, so the pads meet the stem before the blade does, bend round it and hold the cut-off part: about 1.1 N on a 4.5 mm stem, slanted so a piece pulled toward the tips digs in. No glue: the dovetail holds them, and a worn pad slides out for a new one.

5 The jaw on its pivot, and the link

The jaw goes in open, from above; one shoulder screw into the cheek's insert; then the link.



- 1× M3 shoulder screw, 4 × 6, from the lug side
- 1× link (printed)
- 1× dowel pin 2 × 6, pressed into the link

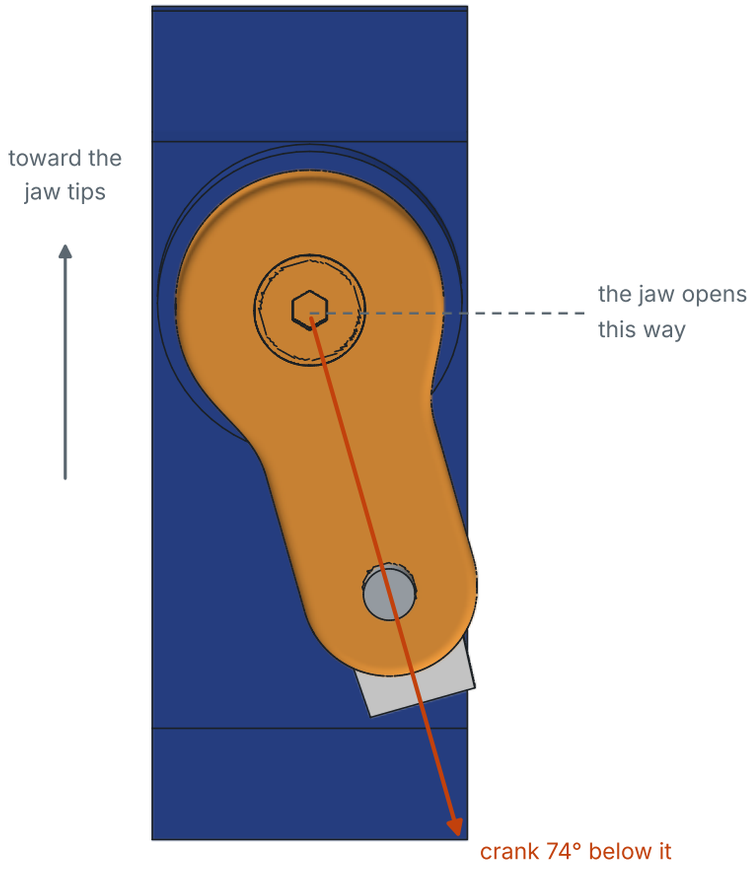
2 mm hex

Hold the jaw open about 30°, lower it into the deck 7 toward its opening side, clear of the fixed pad, then slide it across until its bore meets the cheek's insert, and drive the screw: its head sinks flush into the lug. Close the jaw. Press the dowel into the link, lower the link 6 to the servo side of its slot, past jaw B's rib, and push it across: the dowel drops into the lug.

Tighten the shoulder screw fully: its shoulder bears on the brass insert, not on the jaw, so the jaw keeps 0.2 of play however tight it is. Swing the jaw by hand from closed to open: no tight spot, and it should fall open under its own weight with the tool upside down.

6 The crank on the servo, at closed

Set the servo to its closed position, then fit the crank where it lands.



1× crank and horn, from step 2

1× M2 × 6 countersunk into the spline

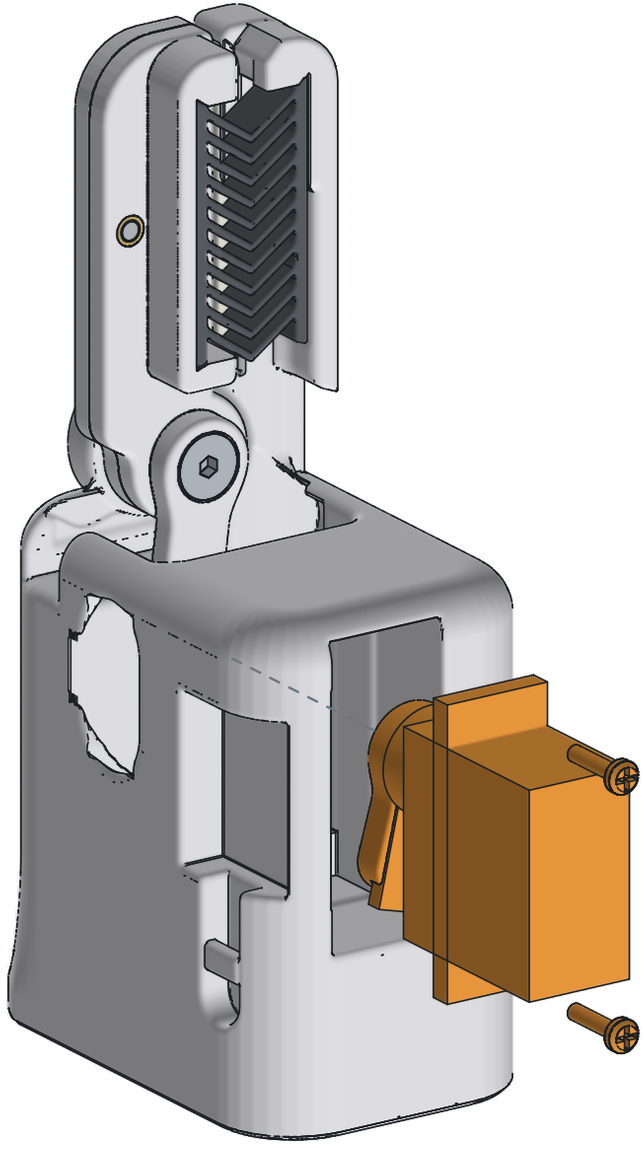
1.5 mm hex · servo tester

Drive the servo to its closed end first: pulse 1500, then 2400 (step 12, or a servo tester). Clockwise seen from the horn means 2400 is closed; if not, use 600. The spline has 20 teeth, 18° each: take the tooth that points the crank nearest the arrow. Software trims the last few degrees.

Seen from the horn side, shaft end toward the jaw tips: the jaw opens to the right. The servo turns the crank 157° from closed to fully open, well inside its 180°. Fit the horn with the crank already over it, then the screw.

7 The servo into the housing

Crank first, from the back; the crank's dowel drops into the link's free end.



- 1× servo with crank, from step 6
- 2× tab screws, from the servo's bag

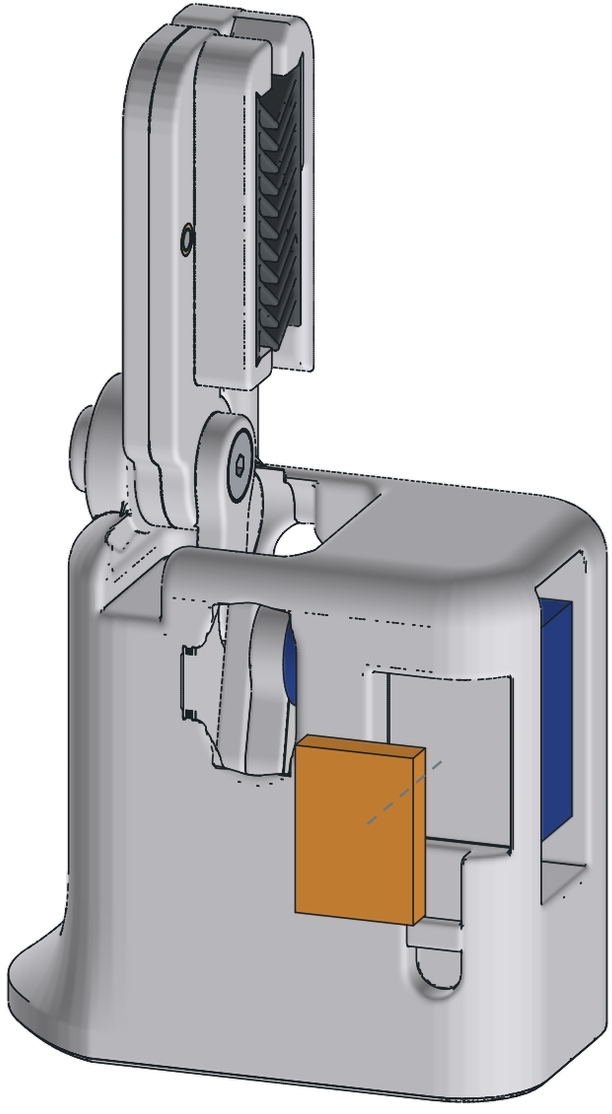
Phillips 1

Hold the link's free end over the crank's dowel through the port on the fixed jaw's side as the servo slides home. The ledge under the tabs is slotted ±1 along the tool: slide the servo until the jaw is fully closed with the crank at its closed angle, then drive the two screws from the back: a driver under 5 thick passes the case.

The jaw is now driven: it should not move by hand with the servo unpowered beyond the gearbox's slack. Leave the servo unplugged until the regulator reads 6.0 V (step 8).

8 Regulator and leads

The servo runs on 6 V from the 12 V wrist line, not on the arm's own supply.



- 1× Pololu D24V10F6, into the bay in the side
- 1× double-sided foam tape, a 12 mm square
- 1× small zip tie, under the bar below the bay

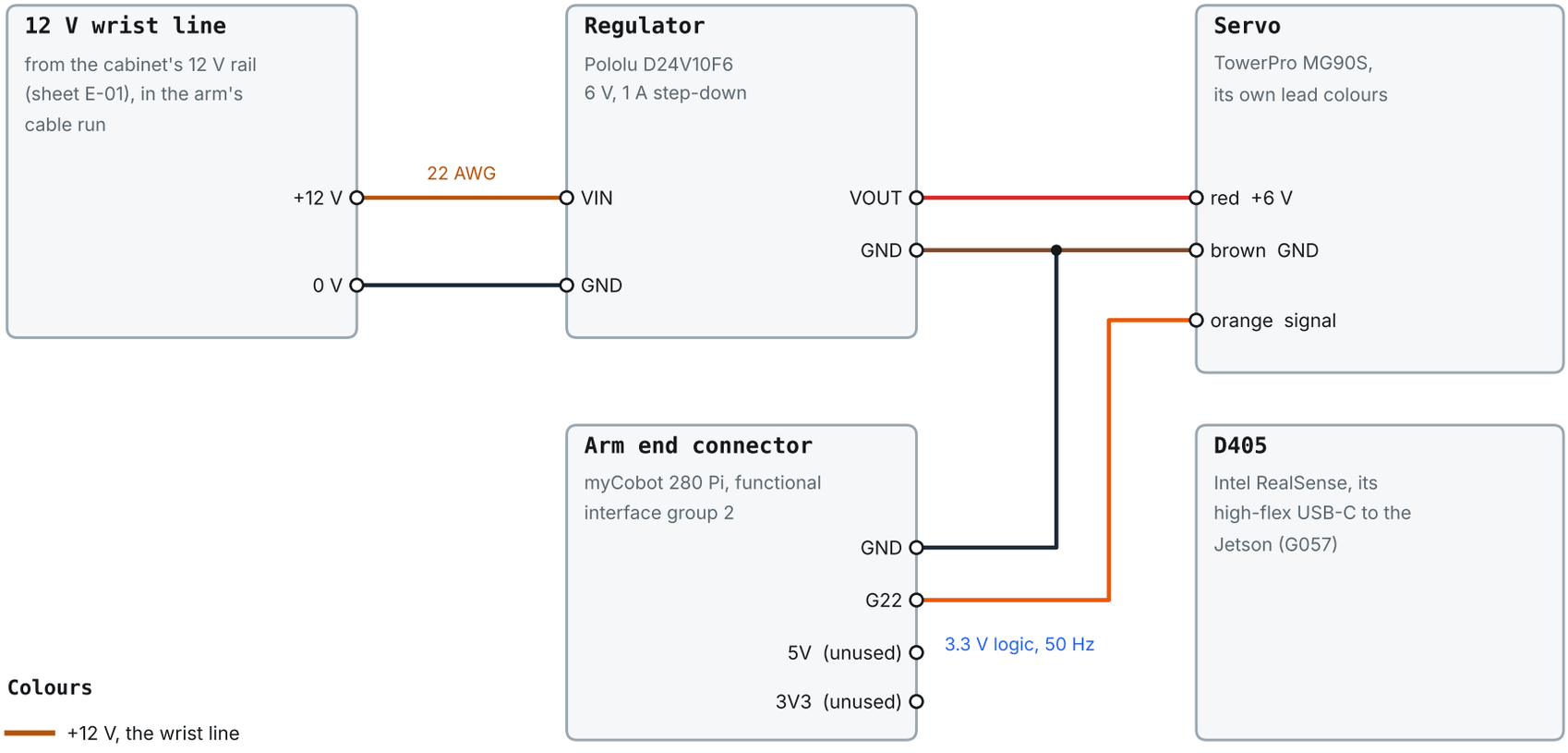
soldering iron · stripper · meter

Solder three wires to VIN, GND and VOUT (the wiring page has which goes where), sleeve the board in heat-shrink, put the tape on its back and press it into the bay, parts outward. The wires leave by the channel at the bay's foot and are tied under the bar with the servo's lead. Power it from the 12 V wrist line and read 6.0 V at VOUT before the servo goes on.

The MG90S stalls at 0.7 A at 6 V; the regulator gives 1 A. The arm's own 5 V pin at the end connector is not rated for that, so only the servo's signal and ground go to the arm.

Wiring · the Herb Snip

Power from the 12 V wrist line, signal from the arm's end connector, video over the D405's USB.



Common ground The regulator's ground and the end connector's GND meet at the servo's brown wire: the signal needs it to mean anything.

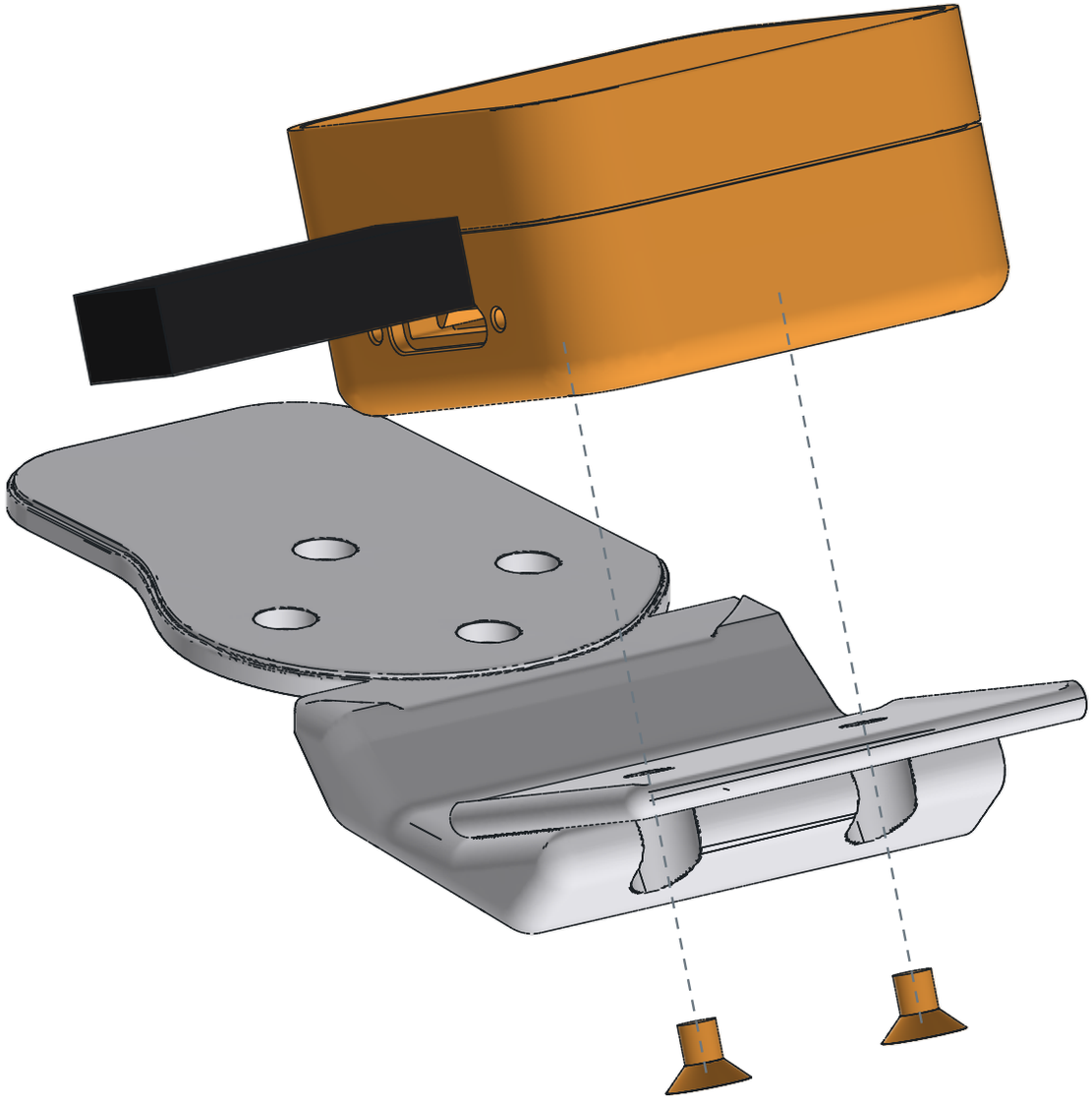
Why not the arm's 5 V The end connector's pins are rated in tens of mA; the servo stalls at 0.7 A.

When the arm is off Doors or E-stop cut the arm's power: no pulses, the servo goes limp, the 6 V stays harmless.

Fallback If the arm will not give a steady 50 Hz on G22, a Pololu Micro Maestro on the D405's USB run drives the same lead.

9 The camera on its ring

On the bench: the D405 you already have, two M3 screws into its back.



1× Intel RealSense D405

2× M3 × 5 countersunk, into the camera's back

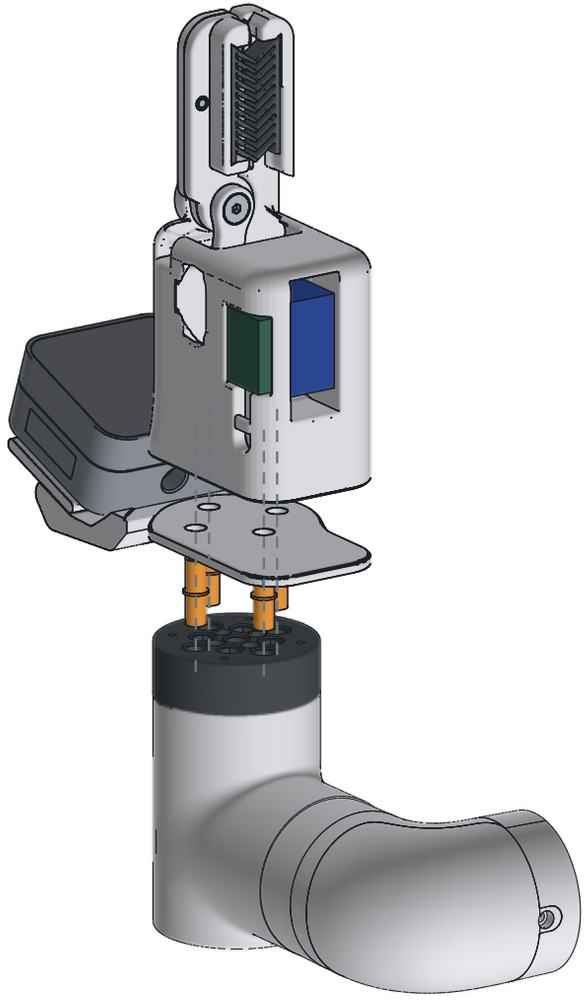
2 mm hex

Before the ring goes on the arm: the screws go in from under the chin, through its two key holes. The camera looks along the jaws, tilted 20° toward the pads; its USB-C socket faces away from the jaw's opening side. Five long, no longer: the heads sink flush in the plate and 2 of thread goes into the camera's shallow holes.

From the camera the cut sits 72 away, just past the D405's 70 mm near limit, and the jaw tips at 90: it sees the stem in the jaws before it cuts. Closer than 70 the depth goes blank, so the last stretch of every approach runs on the arm's kinematics.

10 Onto the arm

Four LEGO pins through the ring into the body. No screws: the 280's end takes accessories this way.



4× LEGO Technic pin with friction

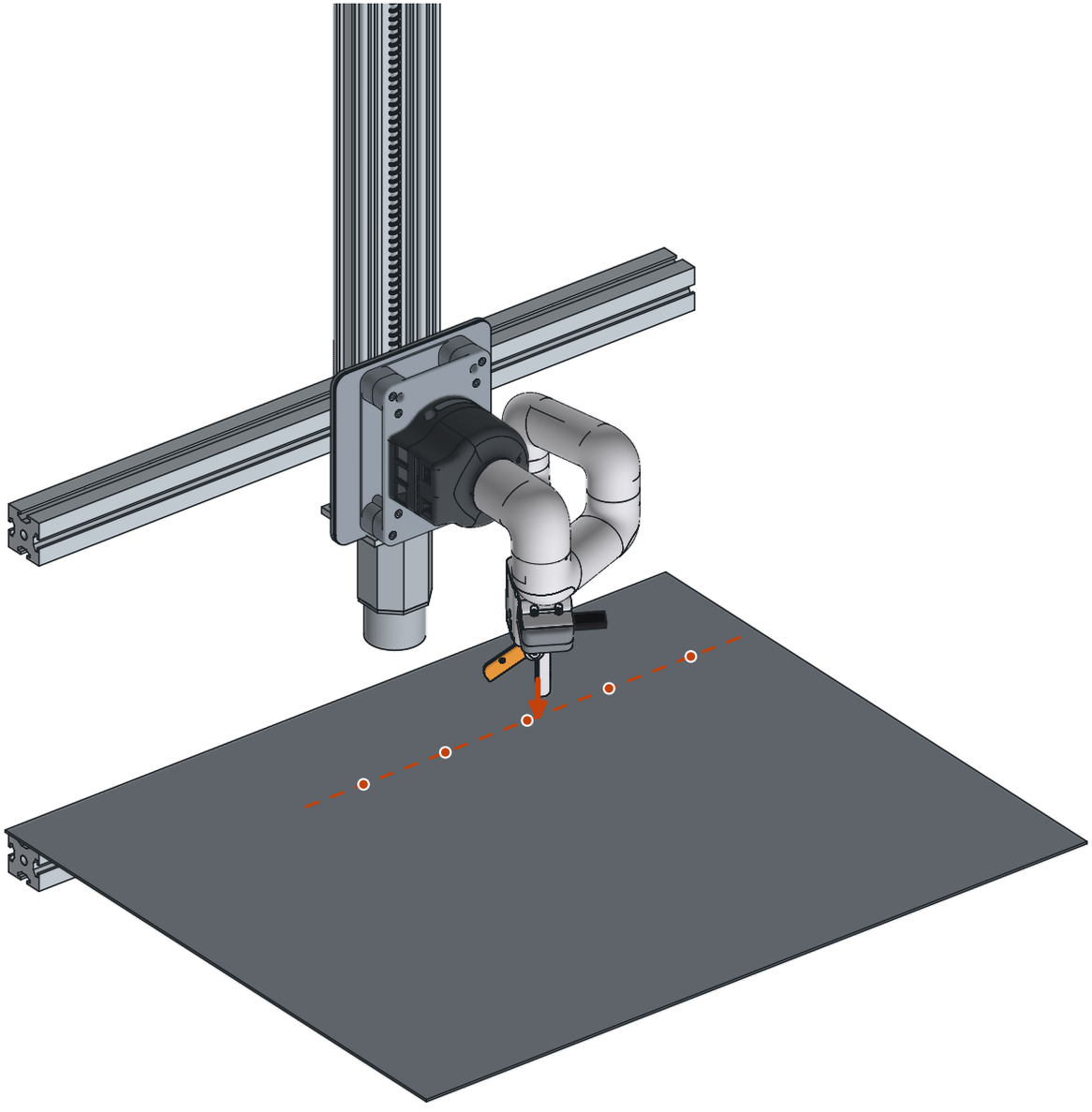
hands only

Take the adaptive gripper off if it is fitted: it pulls off its pins the same way. Pins into the flange first, half their length. Then the ring with its camera over them, flat on the flange, and the Herb Snip square onto them, servo side toward the arm's J6 cable exit: the band's groove runs along the line between them. The pins hold both.

Cables: the D405's USB-C, the 12 V pair and the 3-wire servo lead run together along the arm, tied at each link, with one loose loop at the wrist so J6 can turn ±180°. The servo's signal and ground go to the end connector's G22 and GND.

11 Where the cuts go

Dead leaves into the dead-leaf tray in the base bay; leaves and veggies out to the feeder at tier B.



1× dead-leaf tray, from Harvest Rev B

1× feeder, from Harvest Rev B

Build both from the Harvest booklet. The tray hangs on its rails in the base bay and slides out of the bottom-left door.

A dead leaf rides folded in the closed jaws. The gantry stops over a drop at the lowest Z, J1 368 up, and the jaws open 49 above the rims, at least 58 from a wall. Z full travel there clears the rims by 30. robot.yaml X of the five drops, from the left limit: 18, 102, 186, 270, 355.

Leaves go to the feeder's front compartment, veggies and peppers to its back one: the arm reaches out through the right panel at tier B with the piece in its jaws, on the drop poses the harvest package solves (drops.yaml).

Drawn with the cabinet's left side, everything in front of the corridor and tier A's tray cut away. The dashed line is the dead-leaf tray's rim line, the dots its five drops.

Nothing cut ever leaves the arm's lane except through the feeder's openings. A dead leaf and a harvest never share a destination: the tray takes what the plants shed, the feeder what you eat.

12 Commission

With the cabinet closed, from the Jetson: tools_snip_check.py, then a real stem.

- [ ] **Find the closed end**
  tools_snip_check.py pulse 1500, then 2400: clockwise seen from the horn means 2400 is closed (step 6).
- [ ] **Closed and open**
  step closed_us until the blade just reaches the anvil, plus one; open_us for 55°. Into arm.yaml, calibrated: true.
- [ ] **PWM from the arm**
  tools_snip_check.py pwm · the jaw opens and closes three times on G22's 50 Hz pulses.
- [ ] **Cut a basil stem**
  tools_snip_check.py cut · clean through, and the leaf still in the pads when it opens.
- [ ] **Cut a chili stem**
  the hardest stem in the cabinet. If it only dents it, check the pivot is free (step 5).
- [ ] **Stem in the jaws**
  tools_snip_check.py stem --empty, then stem: empty reads empty, a stem reads present.
- [ ] **Transit fold**
  fold, then run Z full travel over the dead-leaf tray: clear of the tray line and of its rims (step 11).
- [ ] **Into the tray and the feeder**
  jog to each drop (the tray's five, the feeder's two) and write it into robot.yaml. A release there lands inside.

Dead leaves go to the tray, leaves and veggies to the feeder, never the other way. With a door or the feeder's lid open, or the E-stop down, the arm has no power, the end connector sends no pulses, and the servo goes limp: the jaw cannot close on a hand.